ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2	
2	**(a)** 20	Add the three sides.
3	**(c)** 36	Side $= 2r$.
4	**(a)** 16	Two diameters.
5	**(b)** 16	Each is half the diagonal.
6	**(a)** 18	Each side is $2r$.
7	**(b)** 70	Base angles are equal.
8	**(d)** 10	Diameter equals the side.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Every chord through the centre has length 2r, and no other chord is longer.
10	**(c)** A is true but R is false.	Radius = 14 / 2 = 7, so A is true. The diameter is twice the radius, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Inside: 2; On: 1 (1); Outside: 1 (1).	1+1
12	$\angle AOB = \angle COD = 35^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 145^\circ$ (1).	1+1
13	Area of square $= 100$ cm^2 (1); Area of AOB $= 25$ cm^2 (one quarter) (1).	1+1
14	Inside: 1; On: 2 (1); Outside: 1 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distance between adjacent centres $= 2r = 6$ cm (1); Side $= 6$ cm (1); Area $= 36$ cm^2 (1).	1+1+1
16	Width of ring $= 5$ cm (1); Inner diameter $= 8$ cm (1); Longest chord of outer circle = its diameter $= 18$ cm (1).	1+1+1
17	BM $= 5$ cm, BN $= 3$ cm (1); Perimeter $= 2(5+3) = 16$ cm (1); Area $= 15$ cm^2 (1).	1+1+1
	OR: Distance between adjacent centres $= 2r = 8$ cm (1); Side $= 8$ cm (1); Area $= 64$ cm^2 (1).	1+1+1
18	Width of ring $= 7$ cm (1); Inner diameter $= 10$ cm (1); Longest chord of outer circle = its diameter $= 24$ cm (1).	1+1+1
	OR: BM $= 7$ cm, BN $= 4$ cm (1); Perimeter $= 2(7+4) = 22$ cm (1); Area $= 28$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2 \times 3 = 6$ cm (1). (ii) $8 - 5 = 3$ cm (1). (iii) The diameter, $2 \times 8 = 16$ cm (1). (iv) P is outside the 3 cm circle, on the 5 cm circle and inside the 8 cm circle (2).	1+1+1+2
20	Draw AB = 16 cm (1); With A as centre and radius 34 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 30 cm (1); Half diagonal OA $= 17$ cm (1); Perimeter $= 2(16+30) = 92$ cm (1).	1+1+1+1+1
	OR: Row length $= 6 \times 8 = 48$ cm (1); First to last centres $= (6-1) \times 8 = 40$ cm (1); Circles in the strip $= 8$ (1); Show the working with diameter (2).	1+1+1+2

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.
- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 100	Side $= 2r$.
2	**(c)** 24	Two diameters.
3	**(a)** 24	Each is half the diagonal.
4	**(a)** 30	Each side is $2r$.
5	**(d)** 55	Base angles are equal.
6	**(b)** 14	Diameter equals the side.
7	**(d)** 2	
8	**(d)** 32	Add the three sides.
9	**(d)** A is false but R is true.	5 > 4, so the point lies outside and A is false. R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but equal sides do not explain the right angles.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle AOB = \angle COD = 55^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 125^\circ$ (1).	1+1
12	Area of square $= 196$ cm^2 (1); Area of AOB $= 49$ cm^2 (one quarter) (1).	1+1
13	Inside: 1; On: 2 (1); Outside: 1 (1).	1+1
14	$\angle AOB = \angle COD = 75^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 105^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distance between adjacent centres $= 2r = 10$ cm (1); Side $= 10$ cm (1); Area $= 100$ cm^2 (1).	1+1+1
16	Width of ring $= 4$ cm (1); Inner diameter $= 12$ cm (1); Longest chord of outer circle = its diameter $= 20$ cm (1).	1+1+1
17	BM $= 8$ cm, BN $= 5$ cm (1); Perimeter $= 2(8+5) = 26$ cm (1); Area $= 40$ cm^2 (1).	1+1+1
	OR: Distance between adjacent centres $= 2r = 12$ cm (1); Side $= 12$ cm (1); Area $= 144$ cm^2 (1).	1+1+1
18	Width of ring $= 8$ cm (1); Inner diameter $= 14$ cm (1); Longest chord of outer circle = its diameter $= 30$ cm (1).	1+1+1
	OR: BM $= 9$ cm, BN $= 6$ cm (1); Perimeter $= 2(9+6) = 30$ cm (1); Area $= 54$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) CD = AB $= 12$ m (1). (ii) AD = BC $= 5$ m (1). (iii) OA $= \frac{1}{2} \times 13 = 6.5$ m (1). (iv) BD = AC $= 13$ m (1); perimeter $= 2 \times (12+5) = 34$ m (1).	1+1+1+2
20	Draw AB = 12 cm (1); With A as centre and radius 20 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 16 cm (1); Half diagonal OA $= 10$ cm (1); Perimeter $= 2(12+16) = 56$ cm (1).	1+1+1+1+1
	OR: Row length $= 8 \times 4 = 32$ cm (1); First to last centres $= (8-1) \times 4 = 28$ cm (1); Circles in the strip $= 10$ (1); Show the working with diameter (2).	1+1+1+2

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.
- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 36	Each is half the diagonal.
2	**(c)** 42	Each side is $2r$.
3	**(a)** 45	Base angles are equal.
4	**(a)** 18	Diameter equals the side.
5	**(a)** 2	
6	**(a)** 36	Add the three sides.
7	**(d)** 400	Side $= 2r$.
8	**(d)** 36	Two diameters.
9	**(a)** Both A and R are true and R is the correct explanation of A.	O is the midpoint of AC, so OA = 10 / 2 = 5 cm.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the second statement describes a different property.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area of square $= 324$ cm^2 (1); Area of AOB $= 81$ cm^2 (one quarter) (1).	1+1
12	Inside: 1; On: 1 (1); Outside: 2 (1).	1+1
13	$\angle AOB = \angle COD = 100^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 80^\circ$ (1).	1+1
14	Area of square $= 484$ cm^2 (1); Area of AOB $= 121$ cm^2 (one quarter) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distance between adjacent centres $= 2r = 14$ cm (1); Side $= 14$ cm (1); Area $= 196$ cm^2 (1).	1+1+1
16	Width of ring $= 5$ cm (1); Inner diameter $= 16$ cm (1); Longest chord of outer circle = its diameter $= 26$ cm (1).	1+1+1
17	BM $= 10$ cm, BN $= 7$ cm (1); Perimeter $= 2(10 + 7) = 34$ cm (1); Area $= 70$ cm^2 (1).	1+1+1
	OR: Distance between adjacent centres $= 2r = 16$ cm (1); Side $= 16$ cm (1); Area $= 256$ cm^2 (1).	1+1+1
18	Width of ring $= 11$ cm (1); Inner diameter $= 6$ cm (1); Longest chord of outer circle = its diameter $= 28$ cm (1).	1+1+1
	OR: BM $= 6$ cm, BN $= 4$ cm (1); Perimeter $= 2(6 + 4) = 20$ cm (1); Area $= 24$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Perpendicular bisector of a line segment (1). (ii) At the midpoint, 4 cm from A (1). (iii) 90° (1). (iv) With an opening of 4 cm or less the arcs from A and B would not cross (1); every point on the bisector is equally far from A and B, so PB = 5 cm (1).	1+1+1+2
20	Draw AB = 18 cm (1); With A as centre and radius 30 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 24 cm (1); Half diagonal OA $= 15$ cm (1); Perimeter $= 2(18 + 24) = 84$ cm (1).	1+1+1+1+1
	OR: Row length $= 10 \times 8 = 80$ cm (1); First to last centres $= (10 - 1) \times 8 = 72$ cm (1); Circles in the strip $= 12$ (1); Show the working with diameter (2).	1+1+1+2

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.
- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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